

Point Cloud Attribute Dependent Compression using Rate-Distortion Optimized Clustering



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Introduction

Point clouds are a fundamental data structure for representing 3D objects and scenes. Due to their large size, efficient compression is crucial for storage and transmission. This work proposes a novel lossy compression scheme for point cloud attributes, which leverages graph-based transforms and rate-distortion optimization.



Figure 1. "Longdress" point cloud from 8iVFBv2 dataset [1].

Proposed Method

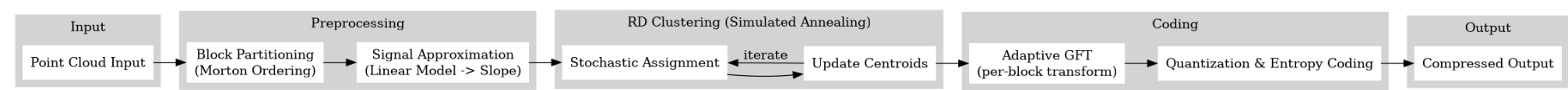


Figure 2. Compression pipeline: Block partitioning → Graph construction → GFT → RD-optimized clustering → Quantization → Entropy coding.

Key Steps:

- Block Partitioning:** Morton ordering preserves spatial locality [5].
- Graph Transform:** GFT concentrates energy in few coefficients [6, 8].
- RD-Optimized Clustering:** Minimize Lagrangian cost: $J = D + \lambda R$ where $\lambda = \alpha \cdot Q^2$ [7].
- Entropy Coding:** RLGR for coefficients [4], Huffman for cluster indices.

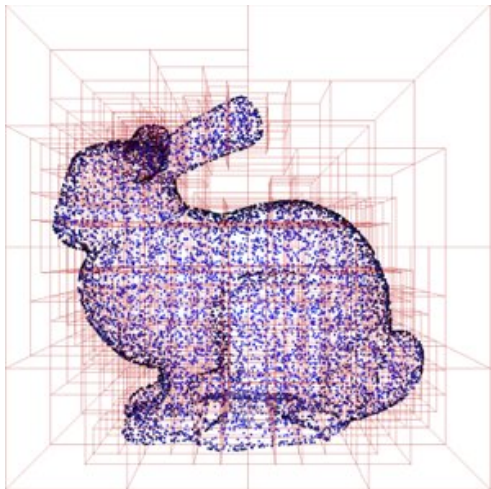


Figure 3. Block partitioning (Morton).

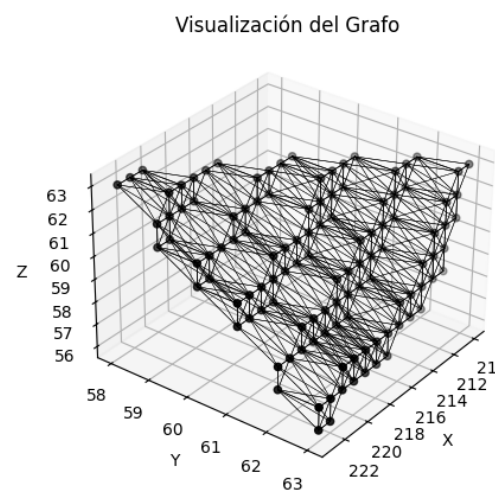


Figure 4. Graph from spatial proximity.

Information Theory Context

Source Model: Non-stationary graph signal with spatial and chromatic correlation [6].

Rate: $R_{\text{total}} = R_{\text{coefficients}} + R_{\text{overhead}}$ where sparsity approximates entropy: $R \approx H(Q(\theta))$ [11].

Distortion: PSNR on luminance (Y) channel from MSE in transform domain.

RD Optimization: Lagrangian finds Pareto-optimal points on achievable (R,D) curve [7].

Adaptive Codebook: Vector quantization creates codebook of K transforms [9]. Trade-off: local adaptation vs. overhead ($\log_2 K$ bits/block).

Key Insight: Smaller blocks (B4) preserve local stationarity, enabling superior adaptation [8].

Experimental Configuration

Dataset: "longdress" frame 1051, 10-bit voxelization [1, 3].

Strategy	Clusters (K)	Iterations	Q-Steps	α
Baseline	1	1	[32, 48, 64]	0.8
RD-Fast	2	10	[32, 48, 64]	0.8
RD-Medium	4	30	[24, 32, 48, 64]	0.8

Table 1. Configuration: $\tau = 0.7$, $B \in \{4, 8, 16\}$.

Huffman Code Analysis

Cluster indices entropy coded using Huffman [10]:

$$L = \sum_{i=1}^K p_i \cdot \ell_i \Rightarrow \text{BPV}_{\text{overhead}} = \frac{L \cdot N_{\text{blocks}}}{N_{\text{voxels}}} \quad (1)$$

L (bits/block): B16 (fast: 1.00, med: 1.95), B8 (fast: 1.00, med: 1.63), B4 (fast: 1.00, med: 1.50).

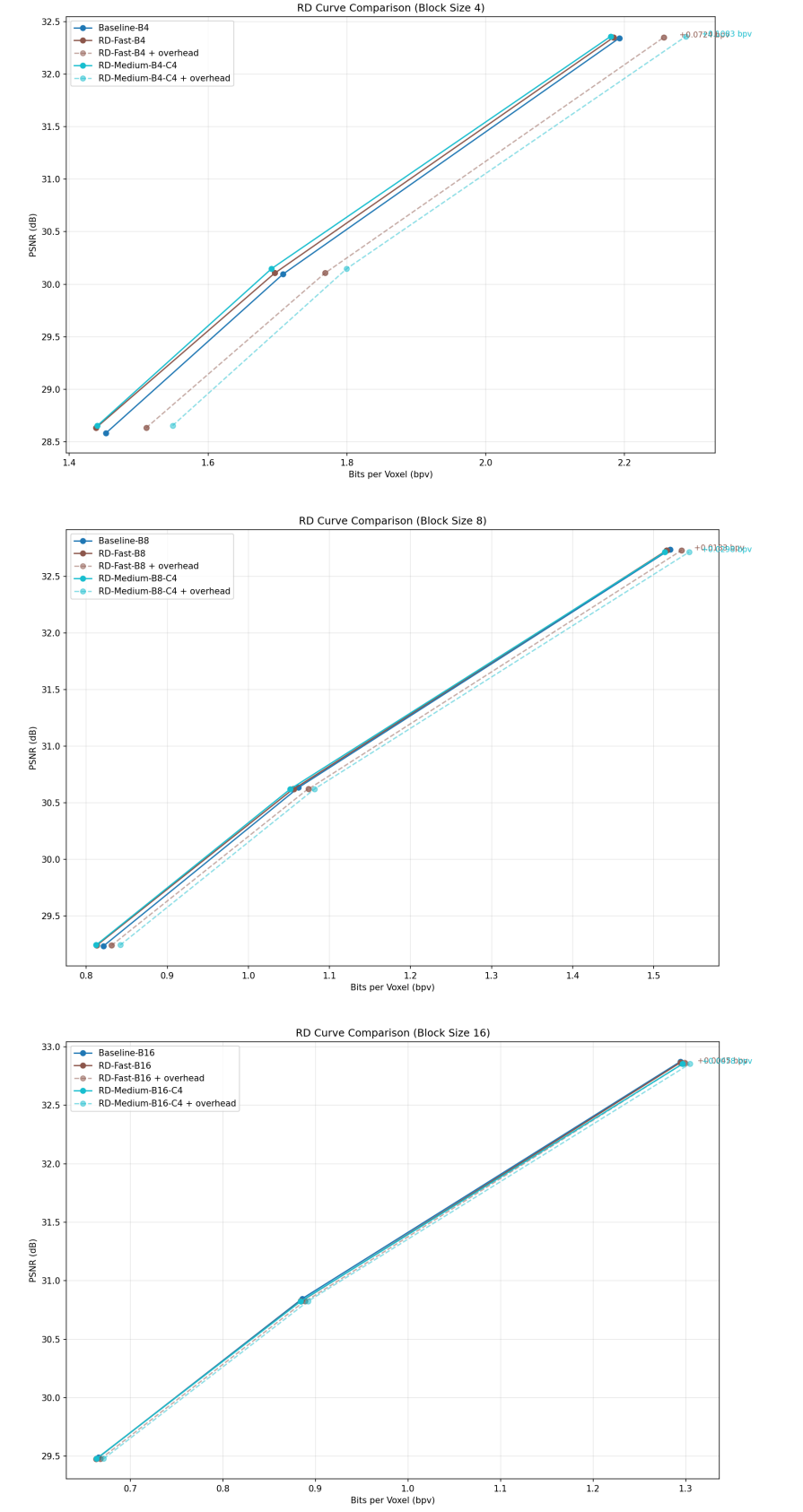
Rate-Distortion Results

Analysis:

The impact of clustering depends strongly on block size.

- B4:** RD-Fast and RD-Medium both outperform the baseline. At 1.44 BPV, RD-Fast gains 0.05 dB with a 1% bitrate drop; RD-Medium reaches 0.07 dB. Across all bitrates, RD-Medium remains slightly superior.
- B8:** Gains are modest. At low bitrate (0.81 BPV) both strategies add 0.01 dB and reduce bitrate by 1%. At higher bitrates, PSNR decreases slightly (−0.01 to −0.02 dB) while still yielding small bitrate savings.
- B16:** Clustering overhead dominates. Both methods show small but consistent PSNR losses (0.01–0.02 dB) with negligible bitrate reduction.

In summary, clustering is clearly effective for **B4**, with consistent gains. Benefits diminish with larger blocks, and RD-Medium offers the best overall tradeoff at the cost of higher complexity.



Performance Analysis: BPV and PSNR Gains

Interpretation: Positive values indicate improvement (bitrate reduction or quality gain).

Method Comparison	Avg. BPV Gain	Avg. PSNR Gain (dB)
Block Size B4		
RD-Fast-B4 vs Baseline-B4	+0.0114	+0.0232
RD-Medium-B4-C4 vs Baseline-B4	+0.0137	+0.0466
Block Size B8		
RD-Fast-B8 vs Baseline-B8	+0.0062	-0.0040
RD-Medium-B8-C4 vs Baseline-B8	+0.0085	-0.0090
Block Size B16		
RD-Fast-B16 vs Baseline-B16	+0.0009	-0.0172
RD-Medium-B16-C4 vs Baseline-B16	+0.0006	-0.0184

Table 2. Comparative performance metrics across configurations.

Key Observations:

- B4:** Both methods achieve bitrate reduction and quality gains; Medium outperforms Fast.
- B8:** Maintains bitrate reduction with minimal quality loss (<0.01 dB).
- B16:** Diminishing returns with minimal bitrate gains and noticeable quality degradation.
- Optimal:** B4 provides best rate-distortion performance for practical use.

Conclusions

Clustering helps only when blocks are small. For B4, both RD-Fast and RD-Medium reduce bitrate and slightly improve PSNR. For B8, gains are minimal but generally neutral. For B16, overhead dominates and performance drops.

Overall, **B4 gives the best tradeoff**, with RD-Medium performing the best.

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